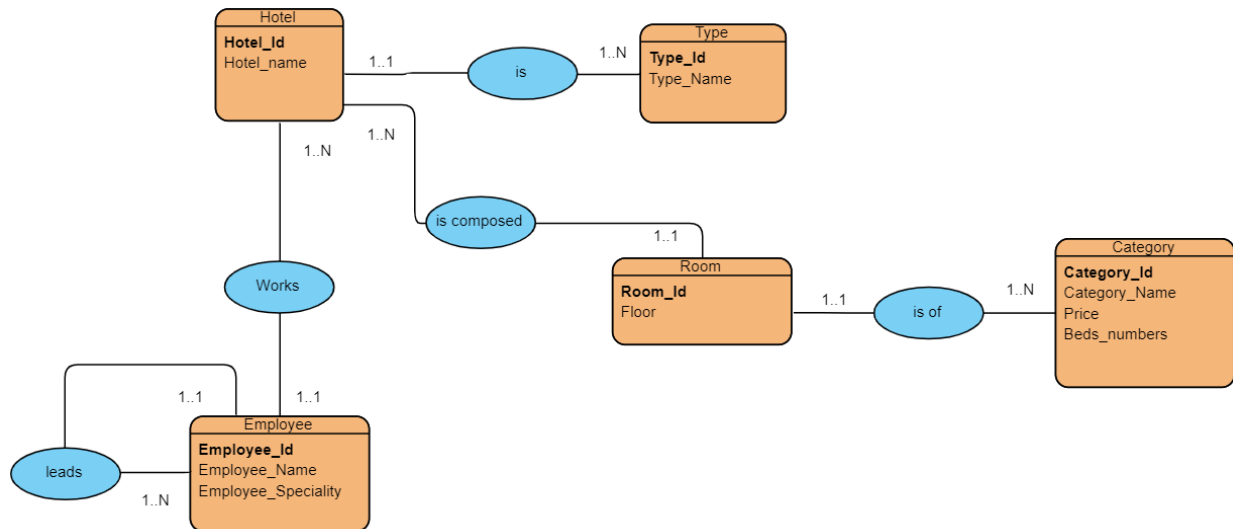


Hotel Management System

Conversion from ER Model to Relational Diagram

Original ER Diagram



Relational Schema

TYPE(Type_Id PRIMARY KEY, Type_Name)

HOTEL(Hotel_Id PRIMARY KEY, Hotel_Name, Type_Id FOREIGN KEY REFERENCES TYPE(Type_Id))

CATEGORY(Category_Id PRIMARY KEY, Category_Name, Price, Beds_Numbers)

ROOM(Room_Id PRIMARY KEY, Floor, Hotel_Id FOREIGN KEY REFERENCES HOTEL(Hotel_Id), Category_Id FOREIGN KEY REFERENCES CATEGORY(Category_Id))

EMPLOYEE(Employee_Id PRIMARY KEY, Employee_Name, Employee_Speciality, Hotel_Id FOREIGN KEY REFERENCES HOTEL(Hotel_Id), Leader_Id FOREIGN KEY REFERENCES EMPLOYEE(Employee_Id))

Relationship Conversion

Relationship	Conversion
Hotel → Type	Type_Id added as foreign key in HOTEL
Hotel → Room	Hotel_Id added as foreign key in ROOM
Room → Category	Category_Id added as foreign key in ROOM
Hotel → Employee	Hotel_Id added as foreign key in EMPLOYEE
Employee leads Employee	Leader_Id self-referencing foreign key in EMPLOYEE

This document presents the conversion of the Hotel Management ER model into a relational schema suitable for database implementation.